

Henry Wilde

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Summary

I am a thoughtful, ethically minded scientist who enjoys building useful things. My work includes public health data science, machine learning, and privacy-preserving engineering. I take pride in learning new tools quickly and applying them to solve real problems.

Core skills

Python data science stack, uv **Version control** git, GitHub, Azure DevOps
Platforming AWS, GCP, Azure, Docker **Automated testing** pytest, hypothesis, unittest.mock, GitHub Actions, Azure Pipelines **Documentation** Diátaxis, Wikis, GitHub Pages, Markdown, LaTeX

Employment

Oct 2024 – present **Research engineer** *Creditsafe*

- Founding member of the R&D team, building tools to improve core products and business processes.
- Technical owner of projects for the Data directorate and Swedish office using technologies including LLMs, knowledge graphs, and data synthesis.
- Mentor to junior engineers, helping to foster a culture of software sustainability and rigour.

LLMs Gemini, OpenAI **Knowledge graphs** Neo4j, TypeDB

Jun 2024 – Oct 2024 **Data scientist** *Dŵr Cymru Welsh Water*

- Collaborated on analytics to support data-driven operational decisions.
- Applied survival analysis to enable proactive maintenance of critical infrastructure.
- Improved software engineering practices through code review and automated testing.

May 2022 – Jun 2024 **Data scientist** *Data Science Campus, Office for National Statistics*

- Led development of an LLM-based reader to summarise ONS activity in parliamentary debates, generating significant cost savings.
- Core developer of the privacy-preserving **record linkage toolkit**, including its secure computation architecture on GCP.
- Mentored a team of apprentices to build a **Python interface** to the 2021 Census API for England and Wales.
- Technical lead for producing **high-fidelity synthetic census microdata** using distributed computing and differential privacy.

LLMs Gemini, OpenAI, LangChain **Publishing** Quarto, Streamlit **Distributed computation** Dask, PySpark, Google BigQuery

Feb 2021 – May 2022 **Research associate** *Water Research Institute, Cardiff University*

- Designed and implemented software infrastructure for the Welsh Government **wastewater surveillance programme**.
- Built reproducible R-based ETL pipelines for biochemical data within one month of self-learning R.
- Developed two core models for monitoring COVID-19 prevalence: a hierarchical GAM for case prediction and a Bayesian dilution model.
- Findings directly informed Welsh Government pandemic policy.

R tidyverse, mgcv, Shiny, RStan, RMarkdown **LIMS**

2019–2020 **Volunteer consultant** *School of Biosciences, Cardiff University*

- Commissioned to improve the school's dissertation allocation process.
- Built a programmatic framework using my open-source **Python matching library**.
- Cut allocation time from a week to seconds while guaranteeing fairness.

Dissertation supervisor *School of Mathematics, Cardiff University*

- Co-supervised an MMORS project on Folk Theorems in game theory.
- Mentored the student on sustainable research software development and report writing.

	2017–2021	PhD studentship teaching <i>School of Mathematics, Cardiff University</i> - Delivered seminars in statistics and computing; supported hackathons and the university maths support service. - Founded an Advanced Python Workshop and code clinic for my peers. - Mentored a high-school student during a Nuffield Research Placement.
Education	2017–2021	PhD Applied Statistics, Operational Research and Data Analytics <i>School of Mathematics, Cardiff University</i> Thesis on the ethical and rigorous use of machine learning in healthcare. - Proposed new perspectives for algorithm evaluation via data synthesis, and fair clustering. - Delivered actionable insights to co-funders using administrative healthcare data. - Accompanied by a suite of open-source research software packages.
	2014–2017	BSc Mathematics (First Class Honours) <i>School of Mathematics, Cardiff University</i> - Explored operational research, computing, and pure mathematics. - Achieved perfect scores in two projects: A&E simulation and iterated Prisoner's Dilemma strategy analysis.
Publications		Full list available on Google Scholar .
Thesis	2021	Wilde, H. <i>New methods for algorithm evaluation and cluster initialisation with applications to healthcare</i> . Cardiff University. PDF . GitHub repository .
Journals	2022	Wilde, H., et al. <i>Accounting for dilution of SARS-CoV-2 in wastewater samples using physico-chemical markers</i> . <i>Water</i> , 14(18):2885. DOI:10.3390/w14182885
	2020	Wilde, H., Knight, V. and Gillard, J. <i>Matching: a Python library for solving matching games</i> . <i>Journal of Open Source Software</i> , 5(48):2169. DOI:10.21105/joss.02169
Awards	2022–2024	Reward and Recognition <i>Office for National Statistics</i> - Eight awards across all three bands, including a sustained excellence award for synthetic data work. - Recognised for accessible technical talks, surge-work contributions, and promoting sustainable software practice.
	2022	PETs Hackathon <i>United Nations PET Lab</i> Finished third out of 200 international teams - Predicted hidden characteristics of Kenyan refugee households using differential privacy within a secure enclave.
Interests		
Cooking		Former chef in fine dining and gastropub kitchens; cooking for friends and family remains my dearest pastime.
Cycling		Restored a vintage steel-frame touring bike during the pandemic and taught myself bike mechanics.
Dungeons & Dragons		Game master for a long-running homebrew campaign with my three brothers; lifelong lover of fantasy and speculative fiction.